RUNNING BITCOIN CORE OR BITCOIN KNOTS

A PRACTICAL GUIDE TO INSTALLING A FULL NODE AND BACKING UP YOUR WALLET ON PAPER, METAL, AND MORE

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A Practical Guide to Installing a Full Node and Backing Up Your Wallet on Paper, Metal, and More

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Contents

Preface

This book is a practical guide for individuals who want to:

- Install and run a Bitcoin full node using Bitcoin Core or Bitcoin Knots.
- Create a wallet for personal funds.
- Back up that wallet safely using a combination of digital, paper, and metal backups.

It is written for non-experts who are comfortable following step-by-step instructions and copying commands, but it includes enough detail to be useful for more technical readers as well.

The focus is on:

- **Security:** Not losing coins and not having them stolen.
- **Simplicity:** A clear, repeatable process suited to personal funds.
- **Self-sovereignty:** You verify Bitcoin yourself, and you control your keys.

This book does not cover trading, speculative strategies, or custodial services. It is about self-custody over your own full node.

Chapter 1

Fundamentals

1.1 What Is a Bitcoin Full Node?

A Bitcoin full node is software that:

- Downloads and verifies every block and transaction according to the Bitcoin protocol rules.
- Relays valid transactions and blocks to other nodes.
- Provides a local view of the network that you can trust without asking anyone else.

Running your own full node gives you:

- **Independent verification:** You do not trust an exchange or wallet app to tell you what is valid.
- **Privacy:** When you query your own node, you do not leak wallet information to a third-party server.
- **Control:** You can enforce your own policy (for example, which features or rules you accept).

1.2 Bitcoin Core vs. Bitcoin Knots

Bitcoin Core is the reference implementation of the Bitcoin protocol. It is the most widely used full node software.

Bitcoin Knots is a derivative of Bitcoin Core maintained by a different developer. It:

- Uses Bitcoin Core as a base.
- Adds extra features and options aimed at power users.
- Is usually compatible with the same blockchain rules, but may have different policy defaults.

For most beginners:

- Bitcoin Core is a perfectly safe and well-supported choice.
- Bitcoin Knots is suitable if you are comfortable with more advanced options and want additional features.

1.3 Keys, Wallets, and Backups

To control Bitcoin, you need **private keys**. A *wallet* is software and/or a file that:

- Stores your private keys.
- Derives addresses where you can receive Bitcoin.
- Signs transactions to spend those coins.

In Bitcoin Core / Knots:

- A wallet is stored as a file (traditionally `wallet.dat`) inside the data directory.
- Modern wallets are HD (hierarchical deterministic), deriving many keys from a single root seed.
- You can encrypt the wallet with a passphrase.

Backups are critical. If you lose your keys and have no backup, *your coins are lost forever.* There is no password reset, no support desk that can recover them.

We will focus on:

- Digital backups (copies of the encrypted wallet file).
- Paper backups (writing recovery material on paper).
- Metal backups (engraving or stamping key material in metal).

1.4 Threat Model for Personal Funds

Before deciding how to back up, think about your threats:

- **Accidental loss:** Hard drive failure, device lost or destroyed, you forget where you stored it.
- **Theft:** Someone physically steals your backup or compromises your computer.
- **Coercion:** Someone forces you to reveal a passphrase.
- **Natural disasters:** Fire, flood, or other events that can destroy paper and electronics.

We will aim for a practical balance:

- For small to medium personal savings, one or two encrypted digital backups plus one physical backup (paper or metal) is usually enough.
- For life-changing money, more redundancy and better physical security are strongly recommended.

Chapter 2

Preparing Your System

2.1 Hardware Requirements

A typical full node requires:

- **Storage:** At least several hundred gigabytes of disk space for the blockchain (preferably SSD).
- **RAM:** 4 GB or more is recommended.
- **CPU:** Any modern 64-bit CPU is fine; initial sync is CPU and disk intensive.
- **Network:** A stable internet connection; several hundred gigabytes of bandwidth during initial sync.

2.2 Operating System Considerations

Bitcoin Core and Knots both support:

- GNU/Linux distributions.
- Windows (64-bit).
- macOS (recent versions).

This book will show example commands mostly using:

- Linux command line.
- Some general notes for Windows and macOS.

2.3 Basic Security Hygiene

Before dealing with money:

- Keep your operating system up to date.
- Use full-disk encryption where possible.

- Use a unique, strong password for your computer.
- Avoid installing random software from untrusted sources.
- Consider having a dedicated machine (or at least a dedicated user account) for your node and wallet.

Chapter 3

Downloading and Verifying Bitcoin Core / Knots

3.1 Where to Download

Important: Always download from official sources and verify signatures. Do not rely on search engine ads or random links.

At the time of writing, official project sites are:

- Bitcoin Core: https://bitcoincore.org/
- Bitcoin Knots: https://bitcoinknots.org/

Navigate to the downloads section appropriate for your operating system.

3.2 Verifying PGP Signatures (Linux Example)

The Bitcoin Core / Knots releases are signed by developers using PGP. Verifying the signature helps ensure that:

- The file was not tampered with in transit.
- You are not installing malware.

Step 1: Download the Binary and Signature

For example (Linux, 64-bit, Bitcoin Core):

```
wget https://bitcoincore.org/bin/bitcoin-core-XX.Y.Z/bitcoin-XX.Y
   .Z-x86_64-linux-gnu.tar.gz
wget https://bitcoincore.org/bin/bitcoin-core-XX.Y.Z/SHA256SUMS.
   asc
```

Replace `XX.Y.Z` with the current version.

Step 2: Import the Release Signing Key

On the official website, you will find a link to the release signing keys. Download and import it:

```
gpg --keyserver keyserver.ubuntu.com --recv-keys <KEYID>
```

Verify the fingerprint against the fingerprint listed on the official site.

Step 3: Verify the Signature and Checksums

Check the signature:

```
gpg --verify SHA256SUMS.asc
```

Then verify checksums:

```
sha256sum bitcoin-XX.Y.Z-x86_64-linux-gnu.tar.gz
```

Ensure the hash matches the entry for your file in `SHA256SUMS.asc`.

3.3 Windows and macOS Verification (High-Level)

On Windows and macOS the idea is the same:

1. Download the installer and the signature or checksums file from the official site.
2. Install GnuPG (or use an existing PGP tool).
3. Import the release signing key and verify the signature.

If you do not feel comfortable with PGP yet:

- At a minimum, verify checksums from the official site.
- Learn PGP signature verification later as a security upgrade.

Chapter 4

Installing Bitcoin Core / Knots

4.1 Linux (Tarball Installation)

After verifying:

```
tar -xzf bitcoin-XX.Y.Z-x86_64-linux-gnu.tar.gz
cd bitcoin-XX.Y.Z/bin
```

You can either:

- Run binaries from this directory, or
- Copy them to a location like `/usr/local/bin`:

```
sudo install -m 0755 -o root -g root -t /usr/local/bin bitcoind
   bitcoin-cli bitcoin-qt
```

4.2 Windows Installation

1. Download the Windows installer (`.exe`) from the official site.
2. Double-click the installer and follow the prompts.
3. Choose a data directory (you can accept the default or select a larger drive).

After installation:

- You can run Bitcoin Core (or Knots) from the Start menu.
- The initial setup wizard will ask about pruning and data directory.

4.3 macOS Installation

1. Download the `.dmg` file.
2. Open it and drag the Bitcoin Core (or Knots) app into `/Applications`.
3. Launch the application; on first run, macOS may warn you about unsigned apps — follow the official instructions for bypassing this if needed.

4.4 Pruned Node vs. Full Archive Node

On first launch, you will be asked if you want a pruned node.

- **Full node (non-pruned):** Keeps the entire blockchain history. Requires more disk space.
- **Pruned node:** Keeps only recent blocks and discards old ones after verifying them. Saves disk space.

For personal use:

- A pruned node is usually fine and much easier on storage.
- Wallet functions work with both.

Chapter 5

Configuring and Running the Node

5.1 Data Directory and Configuration File

The data directory stores:

- Blockchain data.
- Wallet files.
- Logs.
- Configuration file (`bitcoin.conf` or `bitcoin-knots.conf`).

Typical default locations:

- Linux: `$HOME/.bitcoin`
- Windows: `%APPDATA%`
  `Bitcoin`
- macOS: `$HOME/Library/Application Support/Bitcoin`

You can create or edit `bitcoin.conf` in that directory. Example:

```
# bitcoin.conf example
server=1
txindex=0
prune=550
rpcuser=youruser
rpcpassword=long_random_password_here
```

5.2 Starting the Node

On Linux:

```
bitcoind -daemon
```

On Windows/macOS you can run the GUI (Bitcoin Qt) or a background daemon depending on your preference.

5.3 Monitoring Sync Progress

Use the GUI or command line:

```
bitcoin-cli getblockchaininfo
```

Look at:

- `blocks` vs `headers` (how far along you are).
- `verificationprogress` (0 to 1).

Initial sync can take many hours or days, depending on hardware and network.

Chapter 6

Creating a Wallet

6.1 Legacy vs. Modern (Descriptor) Wallets

Modern Bitcoin Core uses descriptor wallets by default. This is good for:

- Cleaner structure of keys and scripts.
- Better compatibility with multisig and advanced setups.

You can create wallets via GUI or CLI.

6.2 Creating a Wallet (GUI)

1. Open Bitcoin Core (or Knots) GUI.
2. Go to **File → Create Wallet**.
3. Choose a wallet name (e.g., "PersonalSavings").
4. Ensure "Encrypt wallet" is checked.
5. Confirm the creation and set a strong passphrase when prompted.

Passphrase rules:

- Make it long (at least 12–16 words or random characters).
- Do not reuse a password from any other site or service.
- Consider writing a hint for yourself (not the whole passphrase) in a separate place.

6.3 Creating a Wallet (CLI)

Example:

```
bitcoin-cli createwallet "PersonalSavings" true
```

The `true` parameter means: create an encrypted wallet. You will be prompted to set a passphrase or need to encrypt afterward, depending on version.

To encrypt an existing wallet:

```
bitcoin-cli encryptwallet -stdinwalletpassphrase
```

6.4 Receiving a Test Transaction

Before putting significant funds:

1. Get a receiving address in the wallet (**Receive** tab in GUI or `getnewaddress` in CLI).
2. Send a tiny amount of BTC from an exchange or another wallet.
3. Wait for confirmations.
4. Practice listing transactions with `listtransactions`.

This ensures everything works and gives you a chance to practice backups with low risk.

Chapter 7

Backing Up the Wallet File

7.1 Understanding `wallet.dat`

The `wallet.dat` file (or descriptor wallet files) contain:

- Your private keys (encrypted if you set a passphrase).
- Metadata about transactions and addresses.
- For descriptor wallets, descriptors that define how keys and addresses are derived.

If your wallet is encrypted and you have a copy of `wallet.dat`, you can restore access to your coins.

7.2 Back Up from the GUI

1. Open your wallet in the GUI.
2. Go to **File** → **Backup Wallet**.
3. Choose a location (USB stick, external drive).
4. Save the backup file (you can rename it, e.g. `personal_wallet_backup.dat`).

Repeat this step for:

- At least one offline USB drive stored safely.
- Optionally, a second backup in another location.

7.3 Back Up from the CLI

Use:

```
bitcoin-cli backupwallet "/path/to/backup/wallet-backup.dat"
```

Make sure:

- The directory exists.
- The backup destination is on a drive you can physically remove and store.

7.4 Storing Digital Backups Safely

- Use **encrypted** external drives when possible.
- Store backups in at least two different physical locations (e.g., home and a trusted relative's place, or a safe deposit box).
- Label backups clearly but not attractively (avoid writing "Bitcoin riches" on the USB stick).
- Periodically plug the backup into a computer to ensure it still reads (without exposing it to untrusted machines).

Chapter 8

Paper and Metal Backups

8.1 Important Warning About Text Backups

Any backup that exposes raw private keys or wallet seed material in human-readable form (paper, metal, photos) can be:

- Stolen by anyone who sees or photographs it.
- Lost if you misplace the paper/metal.
- Coercively extracted.

Never store unencrypted digital copies (e.g., photos) of your seed or private keys in the cloud, email, or on your phone.

8.2 Approach 1: Seed Phrase from an External Wallet

Bitcoin Core itself does not present a simple BIP39 seed phrase in the GUI. One common pattern is:

- Use a hardware wallet or BIP39-compatible software wallet to generate a seed phrase.
- Connect that wallet to your Bitcoin Core / Knots node for transaction broadcasting and verification.
- Write the seed phrase on paper or metal and store it securely.

This combines:

- Full-node verification.
- Simple human-readable backup (seed phrase).

8.3 Approach 2: Descriptor and Key Export (Advanced)

For advanced users, Bitcoin Core supports exporting descriptors or keys. This can be used to create a paper or metal backup.

Examples (do not run blindly, understand the privacy implications):

```
bitcoin-cli getwalletdescriptorinfo "$(bitcoin-cli
   listdescriptors false)"
```

or

```
bitcoin-cli dumpwallet "/path/to/dump.txt"
```

Warning: `dumpwallet` exposes all private keys in plaintext. Treat the resulting file as extremely sensitive and delete it securely once you have transferred the information you need.

If you choose to go this route:

- Use an offline machine if possible.
- Never leave the dump file on an internet-connected device longer than necessary.
- Carefully transcribe required keys or descriptors onto paper or metal.

8.4 Making a Paper Backup

For a seed phrase (from a hardware or BIP39 wallet):

1. Write each word clearly in order.
2. Use a pen that does not smudge easily.
3. Consider writing a second copy and storing it in a different location.
4. Optionally add a checksum or small mark so you can detect if pages are missing.

For descriptors or keys (advanced):

- Print or write them clearly.
- Prefer short, structured data (like descriptors) over long lists of keys.

8.5 Making a Metal Backup

Metal backups are designed to survive:

- Fire.
- Water damage.
- Physical degradation that would destroy paper.

Typical methods:

- Letter punch set and steel plates.
- Commercial metal backup kits that support placing letters for BIP39 words.

Guidelines:

- Work in a private area away from cameras.
- Double-check each word or character before finalizing.
- Store metal backups in a safe, lockbox, or other secure location.

Chapter 9

Restoring From Backups

9.1 Restoring from `wallet.dat`

Scenario: Your computer dies, but you still have your encrypted wallet backup.

1. Install Bitcoin Core or Knots on a new machine.
2. Let it sync (or at least sync up to a point where your wallet's transactions can be seen).
3. Close the node software completely.
4. Copy your backup file into the data directory's `wallets` folder (or main directory for older setups).
5. Start the node.
6. In the GUI, go to **File** → **Open Wallet** and select your restored wallet.
7. Unlock the wallet with your passphrase.

9.2 Restoring from a Seed Phrase (External Wallet)

If you used a BIP39-compatible hardware or software wallet:

1. On a new device, install the same wallet software (or another compatible wallet).
2. Choose "Restore from seed phrase".
3. Enter your 12/24 words in the correct order.
4. The wallet will recreate keys and addresses.
5. Connect this wallet again to your Bitcoin Core / Knots node if desired.

9.3 Testing Your Backups

You should periodically test backup procedures with low amounts:

1. Create a small “test wallet”.
2. Fund it with a tiny amount of BTC.
3. Back it up using your chosen method.
4. Wipe the wallet from your computer (or use a separate device).
5. Restore from the backup and confirm you can access the coins.

Do this before relying on any method for life-changing funds.

Chapter 10

Operational Best Practices

10.1 Separating Hot and Cold Funds

Consider:

- **Hot wallet:** Small amount for spending; may be on a device connected to the internet regularly.
- **Cold wallet:** Larger savings; keys never touch an internet-connected device (hardware wallet or offline setup).

10.2 Passphrase Management

- Use a **password manager** for random, strong passwords for everything except the wallet itself.
- For wallet encryption, many prefer a passphrase they can remember (with written hints stored separately).
- Never reuse wallet passphrases on websites or other services.

10.3 Privacy Considerations

- Running your own node improves privacy, but:
- Avoid reusing addresses.
- Be cautious about linking your identity to your addresses (for example by posting them publicly).

10.4 Keeping Software Up to Date

- Update Bitcoin Core / Knots periodically.
- Before updating, verify signatures as you did initially.
- Always keep backups that were made *before* the upgrade, in case a bug or migration issue occurs.

Chapter 11

Checklists

11.1 Initial Setup Checklist

1. Download Bitcoin Core or Knots from the official site.
2. Verify PGP signatures and checksums.
3. Install on your computer.
4. Configure `bitcoin.conf` with your desired options (pruning, data directory).
5. Start the node and wait for sync.
6. Create an encrypted wallet with a strong passphrase.
7. Make at least two digital backups of the wallet file on separate removable media.

11.2 Backup Checklist

- ☐ Wallet is encrypted with a strong, unique passphrase.
- ☐ At least one encrypted digital backup stored offline.
- ☐ Second encrypted digital backup stored in a different location.
- ☐ Optional: paper or metal backup of seed phrase (if using BIP39/hardware wallet).
- ☐ Tested restore procedure with small funds.

11.3 Periodic Maintenance Checklist

- ☐ Check that node is syncing correctly.
- ☐ Confirm external backup drives still work.
- ☐ Review physical backup locations for signs of damage or risk.
- ☐ Update node software when new stable versions are released, after verification.

Conclusion

Running your own Bitcoin full node and backing up your wallet properly is one of the most empowering things you can do in the digital age. It lets you:

- Verify your own money.
- Control your own keys.
- Reduce reliance on trusted third parties.

With careful backups on digital media, paper, and metal, you can protect your savings against both accidents and attacks. Start with small amounts, practice the full cycle of backup and restore, and gradually grow your confidence and your self-sovereignty.